

Total Points: 50

Submission Due: Sunday, December 23, 2018

Instructions: Please Read Carefully!

- This is an **individual** assignment. Everyone is expected to complete the given assignment on their own, without seeking any help from any website or any other individual. There will be strict penalties for any work found copied from any source and the university policy on plagiarism will be strictly enforced.
- There will be a quiz and/or lab test assessment(s) in the following week based on this assignment.
- You are expected to submit this assignment as:
 - a. Create a single .java file solution of each assignment question. The names of the .java files should be as mentioned in the question.
 - b. Then, create a single .zip file containing the complete solution (.java and .jpg files only) of your implementation. This zip file must be named as ***“Your ID.zip”, for example, “1313001.zip”, and nothing else!***
 - c. The zip file must not contain any other files except the .java and .jpg files.
- Assignment is to be submitted via **Schoology** within the stated deadline above.
 - a. Attach the zip file to the email.
 - b. Write your ID, name, and section in the body of the email.
 - c. Make sure that the email has been successfully delivered and no delivery failure message is received by you.
- Any assignment submission not following the above directions **will not be accepted!**
- No late submissions will be accepted!

[25]

Program Name: Date.java

1. Draw a flowchart and then write a program that inputs three integer values for the **day**, **month**, and **year**. It then displays the message whether the date is valid or invalid.

NOTE: You can either draw a flowchart on paper, and then scan it as a picture, or draw it using the Dia tool. Attach the picture (.jpg) of the flowchart with your .java files.

For example:

```
Enter a value for the day: 10
Enter a value for the month: 12
Enter a value for the year: 2017
The entered date 10-12-2017 is valid.
```

```
Enter a value for the day: 30
Enter a value for the month: 2
Enter a value for the year: 2017
The entered date 30-2-2017 is invalid.
```

This program must follow the following rules:

- i. No negative or zero values can be entered for any of the three values. If a negative or zero value is entered, the program should display that **Invalid value has been entered** and stop.
- ii. The correct value for a **year** should be from 1970 to 2018. Other values are **invalid**.
- iii. There are seven months in which there are a maximum of 31 days. Which months are they? In those months, a **day** value can be a maximum of **31**.
- iv. There are four months in which there are a maximum of 30 days. Which months are they? In those months, a **day** value can be a maximum of **30**.
- v. Month **2** (February) is a special month. If the **year** is a **leap year**, then there are a maximum of 29 days in February. Otherwise, there are a maximum of 28 days.
- vi. A **year** is a **leap year** if it's value when divided by **4** leaves no remainder.
- vii. Use a **Scanner** object to input values.
- viii. Use **Logical operators** for making conditions.

[25]

Program Name: IdentityCardInformation.java

1. Your **CNIC** (Computerized National Identification Card) contains a lot more information than you think. The unique 13 digits identification number is recognized all over the country. There is information in 13 digits. The pattern of the 13 digits identification number is `xxxxx-xxxxxxx-x`. For this assignment, we are only interested in the first part (two digits) and the last part. The first five digits represents unique characteristics. The very first digit of the five digits represents a cardholder's province. See the following list:
 - a. Anyone whose CNIC number starts with the digit **1** shows that they belong to **Khyber Pakhtunkhwa (K-P)**.
 - b. Anyone whose CNIC number starts with the digit **2** shows that they belong to **Federally Administered Tribal Areas (FATA)**.
 - c. Anyone whose CNIC number starts with the digit **3** shows that they belong to **Punjab**.
 - d. Anyone whose CNIC number starts with the digit **4** shows that they belong to **Sindh**.
 - e. Anyone whose CNIC number starts with the digit **5** shows that they belong to **Balochistan**.
 - f. Anyone whose CNIC number starts with the digit **6** shows that they belong to **Islamabad Capital Territory**.
 - g. Anyone whose CNIC number starts with the digit **7** shows that they belong to **Gilgit-Balistan**.

The second digit of the five digits represents the division, third represents the district, fourth the tehsil and the fifth and the last one shows the union counsel in which the cardholder resides. **Gujranwala** is a division and its division digit is **4**. For example, a CNIC number starting with digits 34 means the person is from province Punjab and division Gujranwala.

The **13th** and the last digit of a CNICs shows the gender of the cardholder. Male cardholders are awarded odd numbers like 1, 3, 5, 7 and 9 while the females are issued even numbers.

You are required to write a Java program that takes the **CNIC** number without dashes as a **String** and then it should tell the following:

- i. The province of card holder.
- ii. Whether the card holder belongs to the division Gujranwala or not
- iii. The gender of card holder.

For Example:

Enter CNIC Number: **1510245478962**

The card holder is from province: Khyber Pakhtunkhwa.

The card holder doesn't belong to division Gujranwala.

The gender of card holder is Female.

Enter CNIC Number: **3410245478961**

The card holder is from province: Punjab.

The card holder belongs to division Gujranwala.

The gender of card holder is Male.

Enter CNIC Number: 341024961

Invalid CNIC Number.

Enter CNIC Number: 941024961

Invalid CNIC Number.

This program must follow the following rules:

- i. The length of CNIC number must be 13. No input will be accepted other than 13 digits string. You may use **length()** method to check length of string. If it is not 13, an error message should be displayed that “**Invalid CNIC number.**”.
- ii. No province number other than 1, 2, 3, 4, 5, 6, 7 are allowed i.e. any CNIC number starting from for example 8 is an invalid CNIC number. An error message should be displayed for this condition also i.e. “**Invalid CNIC number.**” You may convert the province digit to `int` for error checking.
- iii. You may use **Scanner** object to get input and display output on console **OR** you may use **JOptionPane** class to take input values and display output.
- iv. Use **Logical Operators** where required.

END OF ASSIGNMENT
